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Assistant Professor Physics Group A

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Assistant Professor Physics Group A

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Question Number : 1 Question Id : 1505611003 Question Type : MCQ Option Shuffling : No

If u and v are conjugate harmonic functions of an analytic function f(z) then _____.

Options :

1.
$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

2.
$$\frac{\partial^2 v}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0$$

3.
$$dv = \left(\frac{\partial u}{\partial x} dy + \frac{\partial u}{\partial y} dx \right)$$

4.
$$dv = \left(\frac{\partial u}{\partial x} dy - \frac{\partial u}{\partial y} dx \right)$$

Which of the following is not a normalized form of a vector ?

Options :

1. $\begin{bmatrix} 1/3 \\ 2/3 \\ 2/3 \end{bmatrix}$

2. $\begin{bmatrix} 3/\sqrt{38} \\ 2/\sqrt{38} \\ -5/\sqrt{38} \end{bmatrix}$

3. $\begin{bmatrix} 1/\sqrt{12} \\ 2/\sqrt{12} \\ 3/\sqrt{12} \end{bmatrix}$

4. $\begin{bmatrix} -2/\sqrt{17} \\ 2/\sqrt{17} \\ -3/\sqrt{17} \end{bmatrix}$

A differential equation is given by $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = \left(\frac{d^2y}{dx^2}\right)^2$ what is its degree ?

Options :

1. 4

2. 6

3. 3

4. 2

If $y_1 = e^{-x} \cos x$, $y_2 = e^{-x} \sin x$ and $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = 0$.

What is the value of Wronskian determinant ?

Options :

1. e^{-x}

2. e^{-2x}

3. $-e^{-2x}$

4. $-e^{-x}$

Question Number : 5 Question Id : 1505611007 Question Type : MCQ Option Shuffling : No

If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$ then inverse of matrix A i.e. A^{-1} is _____.

Options :

1. $\begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$

2. $\begin{bmatrix} -1 & 1 & 0 \\ 2 & -3 & 4 \\ 2 & -3 & 3 \end{bmatrix}$

3. $\begin{bmatrix} 1 & 0 & -1 \\ -2 & -3 & 4 \\ -2 & 3 & -3 \end{bmatrix}$

4. $\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$

Question Number : 6 Question Id : 1505611008 Question Type : MCQ Option Shuffling : No

If $Z_1 = 2 + \frac{3}{2}i$, $Z_2 = -5 + \frac{7}{4}i$, $Z_3 = \frac{5}{4} - \frac{8}{3}i$, $Z_4 = -\frac{11}{2} - i$, then the value of $Z_1 + Z_2 + Z_3 + Z_4$ is _____.

Options :

1. $\frac{-29}{4} - \frac{5}{12}i$

2. $\frac{-29}{4} + \frac{5}{12}i$

3. $\frac{29}{4} - \frac{5}{12}i$

4. $\frac{-5}{12} - \frac{29}{4}i$

Question Number : 7 Question Id : 1505611009 Question Type : MCQ Option Shuffling : No

A square matrix A is called symmetric if _____.

Options :

1. $A = -A^T$
2. $A = A^T$
3. $A^T = 0$
4. $A = I$

Question Number : 8 Question Id : 1505611010 Question Type : MCQ Option Shuffling : No

The function $f(x) = (\sin x)x^3$ in the range $-\pi < x < \pi$ is _____.

Options :

1. an odd function
2. an even function
3. may be even or odd function
4. a numeric

Question Number : 9 Question Id : 1505611011 Question Type : MCQ Option Shuffling : No

The second order linear differential equation of the form $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2)y = 0$ where n-is constant, is called _____.

Options :

1. Bessel's differential equation
2. Legendre's differential equation
3. Laguerr's differential equation
4. Poisson's equation

Question Number : 10 Question Id : 1505611012 Question Type : MCQ Option Shuffling : No

The eigen values of matrix $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ are _____.

Options :

1. $\pm e^{i\theta}$
2. $e^{\pm i\theta}$
3. $\cos \theta \pm \sin \theta$
4. $\pm \tan \theta$

Frame S' is moving with speed v along x-axis, relative to the frame S. A rod is stationary in frame S with length l along x-axis. The length as observed in frame S' is _____.

Options :

1. $\frac{l}{\sqrt{1 - \frac{v^2}{c^2}}}$

2. $l\sqrt{1 - \frac{v^2}{c^2}}$

3. l

4. $\frac{l}{v}$

Which of the following is Poisson bracket of two functions u, v with respect to the canonical variables (q, p) ?

Options :

1. $[u, v]_{q, p} = \frac{\partial u}{\partial q_i} \frac{\partial v}{\partial p_i} - \frac{\partial u}{\partial p_i} \frac{\partial v}{\partial q_i}$

2. $[u, v]_{q, p} = \frac{\partial u}{\partial \dot{q}_i} \frac{\partial v}{\partial p_i} - \frac{\partial u}{\partial \dot{p}_i} \frac{\partial v}{\partial q_i}$

3. $[u, v]_{q, p} = \frac{\partial u}{\partial \dot{q}_i} \frac{\partial v}{\partial \dot{p}_i} - \frac{\partial u}{\partial \dot{p}_i} \frac{\partial v}{\partial \dot{q}_i}$

4. $[u, v]_{q, p} = \frac{\partial u}{\partial q_i} \frac{\partial v}{\partial \dot{p}_i} - \frac{\partial u}{\partial p_i} \frac{\partial v}{\partial \dot{q}_i}$

Check true or false statements.

- A. According to Kepler's law the planets describe circular orbits.
- B. The square of period of revolutions (T^2) are proportional to the cubes of average distances (R^3) of the planets ($T^2 \propto R^3$) from the sun.
- C. For orbital motion centripetal force should act on the planets.
- D. This centripetal force is provided by gravitational attraction of the sun.

Options :

- 1. A-True B-True C-True D-True
- 2. A-False B-False C-False D-True
- 3. A-True B-False C-True D-False
- 4. A-False B-True C-True D-True

Which is the expression for rest energy in special relativity ?

Options :

- 1. $\frac{1}{2} kx^2$
- 2. mgh
- 3. $\frac{1}{2} mv^2$
- 4. mc^2

Choose correct sentences.

- A. Canonical transformation are transformation from old coordinates $\{q\}$ and old momenta $\{p\}$ to new coordinates $\{Q\}$ and new momenta $\{P\}$.
- B. The transformation where new coordinates are functions of old coordinates are called "contact transformation".
- C. Poisson bracket is invariant under the canonical transformation.

Options :

- 1. All sentences are correct
- 2. Only A statement is correct
- 3. Only C statement is correct
- 4. A and B statements are correct

Consider a particle moving in a spiral orbit $r = k\theta^2$, $k = \text{constant}$. What is the law of force for this particle ?

(All symbols have usual meaning)

Options :

1.
$$F(r) = \frac{-L^2}{m} \left\{ \frac{1+k^2}{r^3} \right\}$$

2.
$$F(r) = \frac{-L^2}{m} \left\{ \frac{6k}{r^4} + \frac{1}{r^3} \right\}$$

3.
$$F(r) = \frac{-2L^2}{m} \frac{k}{r^3}$$

4.
$$F(r) = \frac{-L^2 k}{mr^2}$$

Choose correct options from given options below.

- A. A general displacement of a rigid body with one point fixed is a rotation about some axis, this is _____ theorem.
- B. The most general displacement of a rigid body is a translation plus rotation of rigid body, this is _____ theorem.
- i. Euler's Theorem
- ii. Chasles' theorem.
- iii. Norton's theorem.
- iv. Kirchoff's theorem.

Options :

1. A-iv B-iii

2. A-i B-iii

3. A-i B-ii

4. A-ii B-iii

A particle of mass m moves in one dimension such that it has the Langrangian

$$L = \frac{m^2 \dot{x}^4}{12} + m \dot{x}^2 v(x) - v^2(x)$$

Where v is differentiable function of x. The equation of motion for the given mass is _____.

Options :

1. $\frac{1}{2}(m\dot{x}^2) + v(x) = 0$

2. $\left(m\ddot{x} + \frac{\partial v}{\partial x}\right) \left(\frac{1}{2}m\dot{x}^2 + v(x)\right) = 0$

3. $m\ddot{x} + \frac{\partial v}{\partial x} = 0$

4. $\frac{1}{2}m\ddot{x}^2 + \frac{\partial v}{\partial x} = 0$

Which statement correctly represents the first postulate of the special theory of Relativity ?

Options :

1. Laws of physics differ in different inertial frames

2. The law of physics are identical in all inertial frames of reference with respective uniform velocity

3. Speed of light depends on the source

4. Only mechanical laws are invariant

In case of symmetric top, the condition on the principle moments of inertia is _____.

Options :

1. $I_1 = I_2 \neq I_3$

2. $I_1 = I_2 = I_3$

3. $I_1 \neq I_2 = I_3$

4. $I_1 \neq I_2 \neq I_3$

Which statement is True ?

- a. If the centre of mass of the system of particles is not stationary, then angular momentum $\bar{L}$ is independent of the choice of origin.
- b. When external force acting on the system of particles is zero, the linear momentum of the centre of mass is conserved.
- c. When external torque acting on system of particle is non-zero, its angular momentum is conserved.
- d. The angular momentum of a system of particles about a general origin O, equal to the angular momentum of the system concentrated at the centre of gravity plus the angular momentum of the system about its centre of mass.

Options :

1. a
2. b
3. c
4. d

Question Number : 22 Question Id : 1505611024 Question Type : MCQ Option Shuffling : No

Calculate the inertia tensor of a system of four points masses 1 gm, 2 gm, 3 gm, 4 gm located at the points (1, 0, 0), (1, 1, 0), (1, 1, 1) and (1, 1, -1) cm.

Options :

1.
$$\begin{bmatrix} 16 & 9 & 1 \\ 9 & 17 & 1 \\ 1 & 1 & 19 \end{bmatrix}$$
2.
$$\begin{bmatrix} 16 & -9 & 1 \\ -9 & 17 & 1 \\ 1 & 1 & 19 \end{bmatrix}$$
3.
$$\begin{bmatrix} -16 & 9 & -1 \\ 9 & 17 & -1 \\ -1 & 1 & -19 \end{bmatrix}$$
4.
$$\begin{bmatrix} 16 & -9 & -1 \\ 9 & -17 & 1 \\ 1 & -1 & -19 \end{bmatrix}$$

Say whether it is True or False.

- A. Accelerating charge radiates.
- B. Whether the particle is accelerating or deaccelerating, angular distribution of the radiation is the same.
- C. This radiation carries of energy at the expenses of particles kinetic energy.

Options :

- 1. All statements are False
- 2. All statements are True
- 3. Only statement A is True
- 4. Both statements A and C are True

Say whether these statements are True or False.

- A. When a high speed electron hits a metal target, it rapidly deaccelerates giving off a “bremsstrahlung” or breaking radiation.
- B. Poynting theorem is the “work energy” theorem of electrodynamics.
- C. Energy per unit time per unit area transported by the fields is called “Poynting vector”.

Options :

- 1. A-False B-False C-False
- 2. A-True B-False C-False
- 3. A-True B-True C-False
- 4. A-True B-True C-True

Charged particle can be confined in a space by application of _____.

Options :

- 1. Electric field alone
- 2. Magnetic field alone
- 3. Both electric and magnetic field
- 4. Cannot say

Check correct sentences.

- A. In the longitudinal waves, the displacement from equilibrium is along the direction of propagation.
- B. Transverse waves occur in two independent states of polarisation.
- C. If K is wave number and λ is wavelength then $K = 2\pi\lambda$.

Options :

- 1. Sentences A and B are correct
- 2. Only sentence A is correct
- 3. Only sentence B is correct
- 4. Only sentence C is correct

Electromagnetic waves consist of fluctuating electric and magnetic fields such that direction of propagation of wave makes an angle of _____ with electric field and magnetic field.

Options :

- 1. 0°
- 2. 45°
- 3. 60°
- 4. 90°

A toroid with 10^5 turns (10 raise to 5 turns) is wound on a 2m diameter tube. A 10A current passes through each turn of toroid. Determine the magnetic field inside the toroid if $\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$.

Options :

- 1. 2 Tesla
- 2. $4\pi \text{ Tesla}$
- 3. 0 Tesla
- 4. 0.2 Tesla

Say True or False.

- A. Electromagnetic wave consists of electric field.
- B. Electromagnetic wave consists of magnetic field.
- C. In an electromagnetic wave the electric and magnetic fields are parallel to the direction of propagation.

Options :

- 1. Only statement A is True
- 2. Only statement B is True
- 3. Only statement C is True
- 4. Statement A and B are True

If a charge is moved against the electric field, the potential energy of the charge _____.

- a. remains unaffected
- b. increases
- c. decreases
- d. increases K.E. of the charge

Complete the sentence with correct option

Options :

- 1. Only a
- 2. Only b
- 3. c and d
- 4. Only d

According to electromagnetic theory, _____ should radiate energy in the form of electromagnetic waves.

- a. Accelerated charges
- b. Charges in rest position
- c. Electric field
- d. Magnetic field

Options :

- 1. Only a
- 2. Both c and d
- 3. Only b
- 4. Both b and c

Which type of waves cannot pass through a hollow waveguide ?

Options :

1. TE
2. TEM
3. TM
4. Both TE and TM

Choose correct sentence.

- A. Stationary charge produces an electric field E in the space around it.
- B. Moving charge generates a magnetic field B in addition to an electric field E .
- C. Magnetic field can be detected by a compass produced in the wire carrying a current.

Options :

1. Only sentence C is correct
2. Only sentence B is correct
3. Only sentence A is correct
4. All the sentences are correct

Power radiated by a point charge is proportional to _____.

Options :

1. Square of the charge
2. Square of the acceleration
3. Both (1) and (2)
4. Acceleration and charge

The continuity equation in the electrodynamics represent _____.

Options :

1. Conservation of mass
2. Conservation of charge
3. Conservation of momentum
4. Conservation of energy

Say true or false statements.

- A. When perturbation depends upon space coordinates only, then its Hamiltonian H , is assumed to be depends on time.
- B. When perturbation depends upon space coordinates only, then energy is conserved.
- C. The theory of time dependent perturbation was developed by scientist Dirac.

Options :

- 1. A-False B-False C-False
- 2. A-True B-False C-True
- 3. A-True B-True C-False
- 4. A-False B-True C-True

A wave equation in quantum mechanics obeys certain conditions. Identify which of the following is not one of such condition.

Options :

- 1. ψ should be finite everywhere
- 2. ψ should be single value at any place
- 3. ψ and $\frac{d\psi}{dr}$ should be continuous function
- 4. Discontinuity in ψ can exist at boundary

Which of the following is Schrodinger's time independent wave equation ?

a. $\frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} (E - V) \psi = 0$

b. $\frac{\partial^2 \psi}{\partial x^2} + \frac{\hbar^2}{2m} (E - V) \psi = 0$

c. $\frac{\partial^2 \psi}{\partial x^2} - \frac{2m}{\hbar^2} (E + V) \psi = 0$

Where all symbols have their usual meaning.

Options :

1. Both b and c
2. Only c
3. Only a
4. Only b

Choose correct statement.

- A. According to de Broglie's, particle possesses both particle and wave nature.
- B. The wavelength associated with matter is given by $\lambda = h\nu$ where λ = wavelength, h = Planck's constant and ν = frequency.
- C. Particle and wave nature was proved by Davisson and Germer experiment.

Options :

1. Statement A is correct only
2. Statement A and C are correct only
3. Statement C is correct only
4. All statements A, B, C are correct

Commutator of L_x with L_y , where L_x , L_y and L_z are orbital angular momentum operators, is given by _____.

Options :

1. $[L_x, L_y] = i\hbar L_z$
2. $[L_x, L_z] = i\hbar L_y$
3. $[L_z, L_x] = i\hbar L_y$
4. All of the above

The Pauli matrices are related to spin operators by _____.

Options :

1. $S = \frac{\hbar}{2} \sigma$
2. $S = \hbar \sigma$
3. $S = \frac{\hbar}{4} \sigma$
4. $S = i \sigma$

What is following relation commonly known as in quantum physics ?

$\Delta x \cdot \Delta P_x \geq \hbar$ where Δx = Error in location of micro particle

ΔP_x = Error in measurement of linear momentum of microparticle

$\hbar = \frac{h}{2\pi}$ and h = Planck's constant

- a. Schrodinger's equation
- b. Einstein's equation
- c. Maxwell's equation
- d. Heisenberg's uncertainty principle

Options :

1. Only c
2. Only d
3. Only b
4. Only a

Identify the wrong statement.

A. The energy of linear harmonic oscillator $E_n = (n + \frac{1}{2}) \hbar \omega$.

B. For $n = 0$ above equation becomes $E_0 = \frac{3}{2} \hbar \omega$.

C. For $n = 0$ the energy is “Zero point energy”.

Options :

1. Only B statement is wrong
2. Only A statement is wrong
3. Only C statement is wrong
4. A and C statements are wrong

One of the following is not Pauli's spin matrix _____.

Options :

1. $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

2. $\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

3. $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

4. $\begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix}$

The WKB approximation is mainly used for _____.

Options :

1. Relativistic particles
2. Rapidly varying potentials
3. Slowly varying potentials
4. Constant potentials

Which of the following is true statement for degeneracy of states ?

- Two or more energy states have same values though principle quantum no. 'n' is same but orbital quantum no. 'l' is different.
- Two or more energy states have same values though orbital quantum no. 'l' is same but principle quantum no. 'n' is different.
- Every energy state has its unique value irrespective of 'n' and 'l' values.

Options :

- Only c
- Only b
- Both b and c
- Only a

Match their commutation relations.

- | | |
|------------------------------|----------------------|
| A. Position (x) and momentum | a. 0 |
| $(P_x) [x, P_x] =$ | |
| B. $[x^n, P_x] =$ | b. $i\hbar P_z$ |
| C. $[L_x, P_x] =$ | c. $ni\hbar x^{n-1}$ |
| D. $[L_x, P_y] =$ | d. $i\hbar$ |

Options :

- A-d B-c C-a D-b
- A-a B-b C-d D-c
- A-a B-d C-b D-c
- A-a B-c C-d D-b

The lowest quantum mechanical energy of a particle confined in a one dimensional box of size L is 3 eV. The energy of the quantum mechanical ground state for a system of three non interacting spin $\frac{1}{2}$ particles is _____.

Options :

1. 18 eV
2. 11 eV
3. 9 eV
4. 8 eV

Which of the following is correct statement of Liouville's theorem ? (All symbols are standard)

Options :

1. $\frac{d\rho}{dt} = \frac{\partial \rho}{\partial t} + [\rho, H]$
2. $\frac{\partial \rho}{\partial t} = \frac{d\rho}{dt} + [\rho, H]$
3. $\frac{d\rho}{dt} = \frac{\partial \rho}{\partial t} - [\rho, H]$
4. $\frac{d\rho}{dt} = -[\rho, H]$

Which of the following best defines a black body ?

Options :

1. A body that reflects all incident radiation
2. A body that absorbs all incident radiation of all wave lengths
3. A body that transmits all incident radiation
4. A body that emits radiations only in the visible region

The inverse of thermodynamic temperature T is given by one of the following partial derivative _____.

Options :

1. $\left(\frac{\partial S}{\partial E}\right)_{N,V}$

2. $\left(\frac{\partial S}{\partial V}\right)_{E,N}$

3. $\left(\frac{\partial G}{\partial E}\right)_{N,V}$

4. $\left(\frac{\partial H}{\partial E}\right)_{N,V}$

The specific heat of solids at high temperature given by Einsteins theory matches with _____.

Options :

1. Experiment

2. Dulong-Petit law

3. Both (1) and (2)

4. None of the above (1) and (2)

A point in three dimensional phase space has volume given by _____.

Options :

1. $\hbar^3$

2. h^3

3. dx^3

4. dp^3

Question Number : 54 Question Id : 1505611056 Question Type : MCQ Option Shuffling : No

Calculate the possible ways of distribution of 3 bosons in 5 energy states.

Options :

1. 35
2. 42
3. 20
4. 30

Question Number : 55 Question Id : 1505611057 Question Type : MCQ Option Shuffling : No

The intrinsic spin of the fermion is given by _____.

Options :

1. One
2. Zero
3. Half
4. Two

Question Number : 56 Question Id : 1505611058 Question Type : MCQ Option Shuffling : No

The phase space density of a micro canonical ensemble does not change with time. This statement is in accordance with which of the following theorems ?

- a. Clausius inequality theorem.
- b. Liouville's theorem.
- c. Equipartition theorem.
- d. Boltzmann Entropy theorem.

Options :

1. Only b
2. Only c
3. Only a and d
4. Only b and d

Which of the following are examples of degenerate fermi gas ?

- a. Electrons in metals.
- b. Neutron stars.
- c. White dwarf stars.
- d. Ultracold atomic gases.

Options :

- 1. both a and b
- 2. a, b and c
- 3. a, b and d
- 4. all of the above

The ratio of specific heat of electrons in a heated aluminium wire at two different temperatures 327°C and 127°C is greater than _____.

- a. 1.25
- b. 1.36
- c. 1.54
- d. 2.51

Options :

- 1. Only a and b
- 2. Only a
- 3. Only a, b and c
- 4. All of the above

For an ideal gas, the specific heat at constant pressure C_p is greater than that at constant volume C_v because _____.

Options :

- 1. Volume remains constant at constant pressure
- 2. Internal energy depends on pressure
- 3. At constant pressure, part of the supplied heat is used to do External work
- 4. No work is done at constant pressure

Question Number : 60 Question Id : 1505611062 Question Type : MCQ Option Shuffling : No

The canonical partition function 'z' of a system in thermal Equilibrium at temperature 'T' is defined as _____.

Options :

1. $z = \sum_i e^{E_i/KT}$

2. $Z = \sum_i e^{-E_i/KT}$

3. $Z = \sum_i K T / E_i$

4. $Z = \pi i E_i$

Question Number : 61 Question Id : 1505611063 Question Type : MCQ Option Shuffling : No

Planck's constant (h) is dimensionally equal to _____.

Options :

1. Product of linear momentum and distance

2. Product of energy and distance

3. Product of force and time

4. Ratio of energy and time

Question Number : 62 Question Id : 1505611064 Question Type : MCQ Option Shuffling : No

If K_{α} radiation of I(Z = 53) has a wavelength of 0.75 Å, what will be the corresponding wavelength of radiation coming out of Co(Z = 27) ?

Options :

1. 2.33 Å

2. 0.1875 Å

3. 1.86 Å

4. 3.00 Å

Which statement is not true for Raman Spectra ?

Options :

1. Raman Spectrum arises due to scattering of light by vibrating molecules.
2. Raman lines are not polarised.
3. Frequencies of Raman lines are determined not by the scatterer but by incident frequencies.
4. Raman lines are weak in intensity.

Check the properties of Laser and say True or False.

- A. Laser waves are exactly in phase with one another.
- B. Lasers always emits white light.
- C. A laser beam diverges more as compared to ordinary light.

Options :

1. A-True B -True C -True
2. A-True B -True C -False
3. A-False B -False C -True
4. A-True B -False C -False

Say True or False statements.

- A. Splitting of spectral lines in presence of magnetic field is known as Zeeman effect.
- B. In the normal Zeeman effect strong magnetic field is used.
- C. In the normal Zeeman effect spectra splitting takes into three components.
- D. In the anomalous Zeeman effect spectra splits into more than three components.

Options :

1. A-False B -False C -False D-True
2. A-True B -True C -True D-False
3. A-False B -True C -False D-True
4. A-True B -True C -True D-True

Say True or False statements.

The atomic spectral lines in the spectrum obey certain shape selection rules

- A. Selection rule for L is $\Delta L = \pm 1$
- B. Selection rule for J is $\Delta J = \pm 2$
- C. Selection rule for S is $\Delta S = 0$

Where L, S, J are angular momentum.

Options :

- 1. Only statement A is True
- 2. Only statement B is True
- 3. All statements are True
- 4. Only statements A and C are True

In elastic collision between molecules in ground state and photon results into _____ in Raman effect.

- a. scattered photon having higher energy than incident photon.
- b. scattered photon with less energy than incident photon.
- c. decrease in wavelength of scattered photon.

Options :

- 1. Only a
- 2. Only b
- 3. Only c
- 4. a and c

Which of the following is used in NMR technique ?

- a. Non uniform electric field
- b. Nuclear spin and magnetic field
- c. X-rays

Options :

- 1. Only b
- 2. Only a
- 3. Only c
- 4. Other than above

Identify the correct statement describing population inversion.

- a. Atoms in lower energy state are more than those in higher energy state.
- b. No. of atoms are same in lower energy state and higher energy state.
- c. Higher energy level has more atoms than those in lower energy level.

Options :

- 1. Only a
- 2. Only c
- 3. Only b
- 4. a and b

Vibrational-rotational spectra are obtained for _____.

- a. Homonuclear diatomic molecules
- b. Heteronuclear diatomic molecules
- c. Molecules having zero dipole moment

Options :

- 1. Only c
- 2. Only b
- 3. Only a
- 4. a and c

The Kramers-Kronig relations express the connection between _____.

Options :

1. Electric field and Magnetic field
2. Real and Imaginary parts of complex response function
3. Temperature and Pressure
4. Time and Space

Which of the following Lattice Crystal system has one four fold rotation axis as characteristic symmetry element ?

Options :

1. Triclinic
2. Monoclinic
3. Orthorhombic
4. Tetragonal

Which is the statement of Bragg's diffraction condition ?

- a. $2d \sin \theta = n\lambda$
- b. $\Delta K = G$
- c. $2K \cdot G = G^2$
- d. $a_1 \cdot \Delta K = 2\pi v_1$

Where symbols have usual meaning.

Options :

1. Only a
2. Only d
3. a and b
4. a, b, c

Question Number : 74 Question Id : 1505611076 Question Type : MCQ Option Shuffling : No

In case of ferromagnetic materials the difference in energies between easy and hard directions is called as _____.

Options :

1. Magnetostatic Energy
2. Domain Wall Energy
3. Exchange Energy
4. Magnetic Anisotropy Energy

Question Number : 75 Question Id : 1505611077 Question Type : MCQ Option Shuffling : No

Which of the following statement is not correct ?

Options :

1. The microscopic origin of piezoelectricity lies in the displacement of ionic charges with the crystal.
2. The piezoelectric effect is often used to convert electric energy into mechanical energy and vice versa.
3. When a crystal is strained, an electric field is produced within the substance. This is referred as piezoelectricity.
4. Crystals with center of inversion exhibits piezoelectric effect.

Question Number : 76 Question Id : 1505611078 Question Type : MCQ Option Shuffling : No

For Cooper pairs which of the following statement is true ?

Options :

1. The binding energy is strongest when the electrons forming the pair have opposite moments and opposite spins.
2. The Cooper pair is not responsible for Josephson effect.
3. The Cooper pair is responsible for single electron tunneling.
4. All of the above are true.

Match the pairs between crystal structure and their packing fraction.

Crystal structure

Packing fraction

- | | |
|------------------------|------------|
| A. Simple cubic | i. 0.74 |
| B. Body centered cubic | ii. 0.68 |
| C. Face centered cubic | iii. 0.524 |

Options :

1. A-i B-ii C-iii
2. A-ii B-iii C-i
3. A-ii B-i C-iii
4. A-iii B-ii C-i

Say True or False.

- A. In electric dipole, equal positive and negative charges are separated by a small distance (dl).
- B. Electric dipoles are vectors.
- C. Electric dipole moment (μ) , $\mu = q \times dl$, where q is charge.

Options :

1. All sentences are true
2. Only sentence A is true
3. Only sentence B is true
4. Only sentence C is true

Choose correct sentence.

- A. The ratio of charge to voltage is defined as capacitance(C).
- B. Capacitance is measured in Farad.
- C. The unit of electric flux density is coulomb/(metre)².

Options :

1. All sentences are correct
2. Only sentence A is correct
3. Only sentence B is correct
4. Sentences A and C are correct

Say True or False.

- A. Metals have high electrical conductivity.
- B. Metals have low thermal conductivity.
- C. Metallic conductors obey Ohm's law.

Options :

- 1. A-True B-True C-True
- 2. A-True B-False C-True
- 3. A-False B-False C-False
- 4. A-False B-True C-True

Check True/False sentences.

- A. The resistivity of certain metals tends to zero at absolute zero degree Kelvin, known as superconductivity.
- B. All the metal superconductors are diamagnetic in nature below critical temperature (T_c).
- C. In superconductors the entropy decreases on cooling below critical temperature (T_c).

Options :

- 1. A-True B-True C-False
- 2. A-True B-True C-True
- 3. A-True B-False C-False
- 4. A-False B-False C-False

Say True or False.

- A. Vibrational energy of crystals of N, atoms according to classical theory is $= 3NKT$.
- B. In above equation K = Boltzman's constant.
- C. In above equation T = absolute temperature.

Options :

- 1. A-False B-False C-False
- 2. A-True B-False C-False
- 3. A-True B-True C-False
- 4. A-True B-True C-True

A magnetic material is subjected to an external magnetic field of 10^5 A/m. Calculate its magnetic susceptibility if magnetization is -5 A/m in it. Also identify the type of magnetic material.

- a. 5×10^{-5} , Paramagnetic
- b. -5×10^{-5} , Diamagnetic
- c. $+5 \times 10^5$, Ferromagnetic
- d. 0.5×10^5 , Ferrimagnetic

Options :

- 1. Only c
- 2. Only a
- 3. Only b
- 4. Only d

In a sequential circuit the output at any instant depends on _____.

Options :

- 1. Present input and past state
- 2. Present input only
- 3. Past input (previous state) only
- 4. None of the above

Question Number : 85 Question Id : 1505611087 Question Type : MCQ Option Shuffling : No

An intrinsic sample of a germanium crystal has a Hole density 10^{13} cm^{-3} at room temperature. When it is doped with antimony, the Hole density decreases to 10^{11} cm^{-3} at same temperature. Find the value of the majority carrier density.

Options :

1. 10^{37} cm^{-3}
2. 10^{-15} cm^{-3}
3. 10^{15} cm^{-3}
4. 10^{-37} cm^{-3}

Question Number : 86 Question Id : 1505611088 Question Type : MCQ Option Shuffling : No

A P-N junction diode can be used as a rectifier because _____.

Options :

1. It conducts in both direction
2. It conducts in only one direction
3. It does not conducts at all
4. It can conduct at zero applied voltage

Question Number : 87 Question Id : 1505611089 Question Type : MCQ Option Shuffling : No

If for a non-inverting OPAMP, input resistance is $1 \text{ k}\Omega$ and feedback resistance is $5 \text{ k}\Omega$, then the voltage gain is given as _____.

Options :

1. 2
2. 6
3. 5
4. 4

Question Number : 88 Question Id : 1505611090 Question Type : MCQ Option Shuffling : No

A certain OPAMP has bias current of $70\ \mu\text{A}$ and $70.15\mu\text{A}$. The input offset current of the OPAMP is _____.

Options :

1. $100\ \text{nA}$
2. $150\ \text{nA}$
3. $1.5\ \mu\text{A}$
4. $0.1\mu\text{A}$

Question Number : 89 Question Id : 1505611091 Question Type : MCQ Option Shuffling : No

Zero biasing is unique for _____.

Options :

1. JFET
2. BJT
3. Enhancement type MOSFET
4. Depletion type MOSFET

Question Number : 90 Question Id : 1505611092 Question Type : MCQ Option Shuffling : No

The gain of an amplifier without feedback is 50. With negative voltage feedback, the gain is reduced to 25. Due to ageing, the amplifier gain without feedback falls to 40. Find the percentage reduction in the stage gain.

Options :

1. 80 percent
2. 30 percent
3. 60 percent
4. 20 percent

A logic diagram of a particular function expressed as

$$O = (A + B + C + D) (A + B + C + \bar{D}) (A + B + \bar{C} + D) \\ (A + B + \bar{C} + \bar{D}) (\bar{A} + \bar{B} + \bar{C} + \bar{D})$$

has which of the following combination of logic gates ?

- a. 5 NOT gates, 5 OR gates and 1 AND gate
- b. 4 NOT gates, 5 OR gates and 1 AND gate
- c. 4 NOT gates, 6 OR gates and 2 AND gates
- d. 5 NOT gates, 3 OR gates and 1 AND gate

Options :

- 1. Only a
- 2. Only b
- 3. Only c and d
- 4. Only b and c

Energy required to break down a nucleus into its component nucleons is called _____

- a. Nuclear binding energy
- b. Electron energy
- c. Atomic energy
- d. Nuclear Breakdown

Complete the definition using correct option.

Options :

- 1. Both b and c
- 2. Only d
- 3. Only a
- 4. Only c

Match the pairs with correct option.

- | | |
|--------------------|-------------------------|
| a. α – rays | i. Negative charge |
| b. β – rays | ii. Neutral (No charge) |
| c. γ – rays | iii. Positive charge |

Options :

1. a-i b-ii c-iii
2. a-ii b-iii c-i
3. a-iii b-i c-ii
4. None

In high-energy electron scattering and muonic x-ray experiments, what is primarily measured ?

Options :

1. Nuclear mass defect
2. Energy of neutrons
3. Distribution of nuclear matter
4. Distribution of nuclear charge

What is meant by nuclear fusion ? It is a process of combining two _____.

- a. Heavy nuclei
- b. One light and one heavy nucleus
- c. Light nuclei
- d. Neutrons

Options :

1. Only a
2. Only c
3. Only b
4. Only d

Which of the following reaction is allowed under the law of conservation of strangeness ?

- a. $\pi^+ + n = \Lambda^0 + k^+$
- b. $\pi^+ + n \rightarrow k^0 + k^+$
- c. $\pi^+ + n \rightarrow \bar{k}^0 + \Sigma^+$
- d. $\pi^- + p \rightarrow \pi^0 + \Lambda^0$

Options :

- 1. Only a
- 2. Only d
- 3. Both b and c
- 4. All are allowed

The so called magic numbers (z = 2, 8, 20, 28, 50, 82 and 126) represents _____.

Options :

- 1. The proton or neutron numbers at which the nucleus shows sudden and discontinuous behaviour in separation energy
- 2. The atomic numbers of inert gases in periodic table
- 3. The energy levels of electrons in an atom
- 4. The nucleon numbers corresponding to unstable nuclei

Which of the following elementary particle is Baryon ?

Options :

- 1. Kaon
- 2. Muon
- 3. Sigma
- 4. Electron

What is name of the last sub-atomic particle discovered in the year 2012 ?

- a. Leptons
- b. Mesons
- c. Higgs Bosons
- d. Hadrons

Options :

1. Only b
2. Only a
3. Only d
4. Only c

Which of the following particle has unit spin ?

Options :

1. Muon
2. Electron-neutrino
3. Photon
4. None of the above